

A Survey for Generative Search Engine Optimization

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Generative Search Engines (GSEs) are fundamentally reshaping the paradigm of information access, shifting users from actively exploring ranked result lists to passively consuming synthesized knowledge embedded within generated responses. Powered by large language models (LLMs) and retrieval-augmented generation (RAG), these systems integrate information from diverse sources to produce comprehensive and context-aware answers. This transformation challenges the adequacy of traditional Search Engine Optimization (SEO), whose optimization objectives are primarily centered on ranking within search result pages and are therefore misaligned with generative retrieval settings. In response, Generative Search Engine Optimization (G-SEO) has emerged as a new research direction, emphasizing the competition for visibility and attention within generated responses rather than external ranking positions. Recent studies have proposed a variety of G-SEO techniques, including prompt injection and semantic rewriting, to influence the selection and presentation of information in generative outputs. Meanwhile, systematic evaluation frameworks, benchmarks, and metrics have been introduced to quantify optimization effectiveness in generative search environments. This survey consolidates the existing body of research on G-SEO, formally defines its fundamental tasks and problem settings, and provides a comprehensive analysis of current challenges and future research directions.

CCS Concepts: • **Information systems** → *Retrieval models and ranking*; *Content ranking*; **Information retrieval**; • **Computing methodologies** → Natural language processing.

Additional Key Words and Phrases: Generative Search Engine Optimization, Retrieval Augmented Generation, Information Retrieval, Content Manipulation, Data Poisoning, Hijacking, Prompt Injection

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1 Introduction

Driven by the rapid advancement of large language models (LLMs) and Retrieval-Augmented Generation (RAG), Generative Search Engines (GSEs) are fundamentally reshaping the paradigm of information access. Unlike traditional search engines, which present ranked lists of documents, GSEs integrate information from multiple sources and generate coherent, context-aware responses in the form of natural language passages. This emerging paradigm shifts users from actively exploring and selecting relevant information from ranked lists to passively consuming synthesized answers

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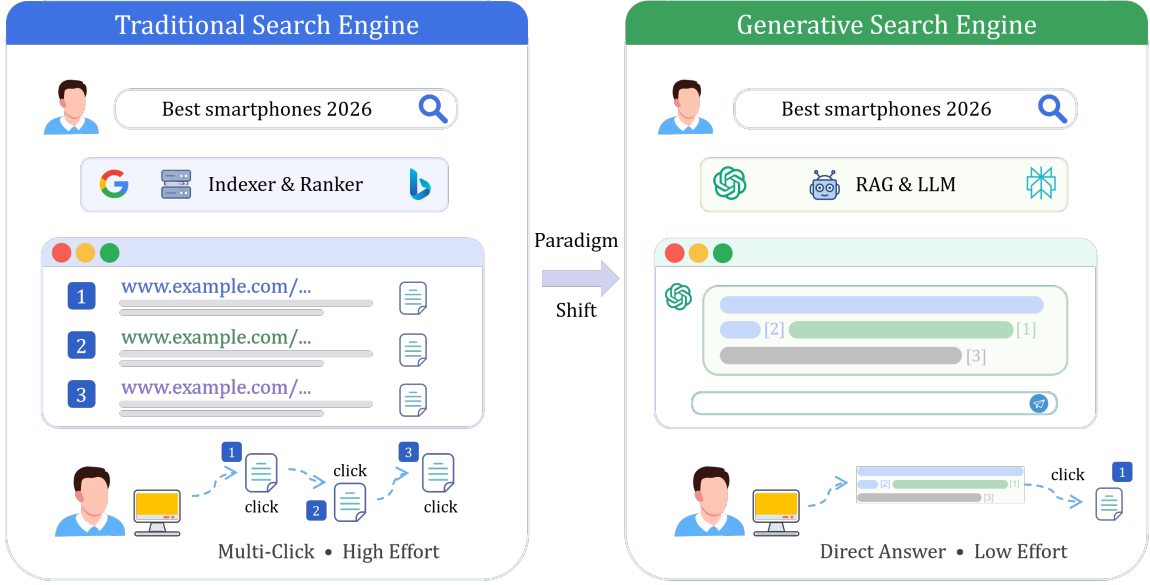


Fig. 1. Paradigm shift from traditional search engines to generative search engines. Traditional search engines return ranked webpages that require multi-click browsing and manual information integration. In contrast, generative search engines use RAG and LLMs to provide direct answers with citations.

directly provided by the system. While this transformation significantly enhances user experience, it also redefines how visibility and influence are established within the digital information ecosystem.

1.1 Background and Motivation

In recent years, the paradigm of information retrieval has undergone a profound transformation [57]. As illustrated in Figure 1, traditional search engines rely on relevance-based ranking mechanisms to organize large-scale web content into ordered lists, leaving users responsible for filtering and integrating information. In contrast, GSEs inherit retrieval capabilities while leveraging the reasoning and comprehension abilities of LLMs, together with RAG techniques, to perform deep semantic analysis, integration, and reconstruction of multi-source content, often accompanied by supporting citations. The resulting responses are coherent, precise, and context-aware. This shift marks a paradigm transition in information access from list-based browsing to content-centric understanding and knowledge synthesis.[12, 61] By focusing on semantic alignment between queries and content sources, GSEs eliminate the need for users to manually interpret and aggregate information. Consequently, users increasingly rely on holistic understanding derived from generated responses rather than sequential browsing. This paradigm significantly reduces irrelevant information exposure, lowers cognitive load, and improves the efficiency and effectiveness of information acquisition.

However, as GSEs become a dominant access channel, they challenge the effectiveness of traditional Search Engine Optimization (SEO). Conventional SEO techniques, which are well-established and highly mature, rely on both on-page and off-page factors [31], including keyword optimization [30, 78], metadata design [24], webpage structure [60], and backlink strategies [2]. These methods are fundamentally grounded in a retrieval paradigm centered on relevance-based ranking and list-based result presentation. Their objective is to improve document visibility by enhancing the alignment

between webpage content and user queries. In generative search settings, however, the optimization objective shifts from achieving higher rankings to ensuring that content is selected, incorporated, and cited within generated responses. The replacement of ranked lists with synthesized answers fundamentally disrupts the underlying assumptions of SEO, requiring a rethinking of optimization strategies.

In this context, Generative Search Engine Optimization (G-SEO) has emerged as a new research direction [1]. G-SEO shifts the optimization focus from ranking positions to visibility and influence within generated responses, emphasizing how content is selected and reflected in the final outputs. This paradigm introduces new challenges, including semantic alignment with model reasoning, interaction with prompt and context representations, and adaptation to complex generation pipelines. Recent studies have proposed various G-SEO techniques, including prompt injection methods [36, 50] and semantic rewriting strategies [25, 53], to influence the selection and presentation of information in generated responses. Meanwhile, retrieval-side manipulation approaches, such as RAG poisoning [39, 85], reveal potential vulnerabilities in generative search pipelines, even without directly modifying original content sources. These studies highlight the necessity of a systematic investigation and synthesis of G-SEO.

1.2 Scope and Contributions

This survey aims to provide a comprehensive and systematic overview of the emerging field of G-SEO. We focus on the interaction between content optimization strategies and generative models, examining how different techniques influence the behavior of GSEs. Specifically, the main contributions of this survey are as follows:

- We formalize the task definition and underlying assumptions of G-SEO, clarifying its distinctions from traditional SEO and related paradigms.
- We propose a unified taxonomy of G-SEO methods, covering both retrieval-side manipulation and generation-side optimization.
- We systematically review existing evaluation methodologies, including metrics, benchmark datasets, and experimental protocols.
- We summarize key challenges, such as long-context scenarios, cross-model generalization, and adversarial robustness, and discuss associated ethical considerations.

1.3 Organization of the Survey

The remainder of this survey is organized as follows. Section 2 introduces the task definition and assumptions of G-SEO. Section 3 presents evaluation methodologies, including metrics, datasets, and experimental setups. Section 4 provides a taxonomy of existing G-SEO methods, distinguishing between retrieval-side and generation-side approaches. Section 5 discusses key challenges and broader implications, including technical limitations and ethical considerations. Finally, Section 6 concludes the survey and outlines future research directions.

2 Task Definition and Assumptions

This section formalizes the task definition and underlying assumptions of G-SEO. We begin by contrasting traditional SEO with G-SEO to highlight the paradigm shift in optimization objectives. We then present a conceptual formulation of G-SEO, followed by a set of system-level assumptions that support subsequent analysis.

2.1 From SEO to G-SEO

Compared to traditional SEO, which is grounded in a relevance-based ranking paradigm, G-SEO operates in a generative retrieval setting centered on multi-source information integration. Its optimization objective is no longer limited to improving webpage rankings within search engine result pages (SERPs), but instead focuses on how content is represented within generated responses [10]. This paradigm shift introduces several fundamental differences. The notion of visibility transitions from document-level exposure to content-level inclusion in generated outputs. Accordingly, optimization strategies extend beyond improving retrieval relevance to influencing the interaction between content and the two core system components, retrieval and generation parts, thereby increasing the overall complexity of the optimization problem. The evaluation of optimization effectiveness also undergoes substantial changes. In addition to traditional ranking-based metrics, such as position [3, 53] and hit rate [35], G-SEO requires generation-oriented evaluation metrics, including citation rate [1, 71] and influence [11]. While the above comparison provides an intuitive understanding of G-SEO, a systematic analysis requires a formal characterization of its optimization process. We therefore proceed to conceptualize G-SEO within a unified retrieval-generation framework.

2.2 G-SEO Conceptual Foundations

At a fundamental level, there is a growing consensus that G-SEO should be formulated as an optimization problem over a coupled retrieval-generation pipeline under a black-box setting [1, 50, 85]. Given a user query, GSEs typically retrieve a set of relevant content candidates and subsequently synthesize them into a final response via a generative model. However, as noted by Marco [46], most existing G-SEO studies adopt a narrow perspective that focuses primarily on the generation stage, while overlooking retrieval-stage manipulations—such as RAG poisoning—that can also significantly influence system behavior. To establish a more comprehensive framework, we extend the scope of optimization to include not only individual content sources but also the retrieved content set itself. This leads to a generalized formulation of G-SEO with two key optimization dimensions: retrieval-level and generation-level optimization.

Formally, let q denote a user query, S the corpus of available content source, and $\mathcal{R}(S, q)$ the retrieval function that selects relevant documents or passages. A generative model $\mathcal{G}$ produces the final response as $r = \mathcal{G}(q, \mathcal{R}(S, q))$. The objective of G-SEO is to optimize existing content source $s^* = \mathcal{O}(s), s \in S$, or construct new content source to extend the corpus as $S^* = S \cup \{s^*\}$, such that s^* and its associated content source are (1) more likely to be selected by the retrieval function, and (2) exert greater influence on the generated output. It is important to note that G-SEO fundamentally operates at the semantic level rather than relying solely on lexical matching. Effective optimization thus requires aligning content source with the latent representations mechanisms of generative retrieval systems.

2.3 G-SEO Models and System Assumptions

The above formulation characterizes G-SEO at an abstract level. To facilitate systematic analysis and comparison across methods, it is necessary to further specify the system assumptions under which G-SEO is studied. As illustrated in Figure 2, we introduce three common assumptions that underpin most existing G-SEO research [1, 71], while remaining consistent with real-world GSE systems [20, 74].

2.3.1 RAG Architecture. The GSE system is assumed to follow a RAG architecture consisting of a retriever and a generator. The retriever selects relevant content source based on the query, and the generator produces the final response conditioned on both the query and retrieved content source.

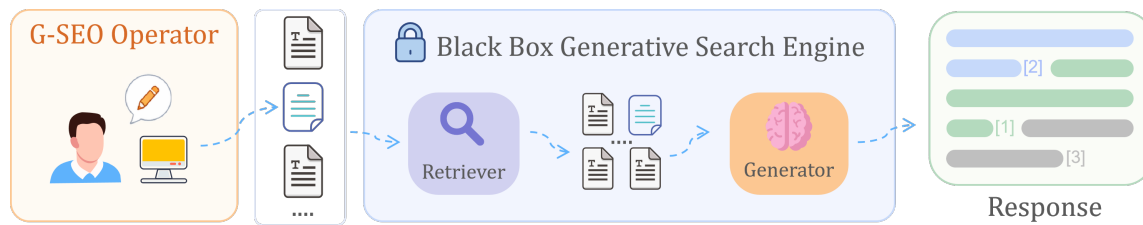


Fig. 2. General framework of G-SEO under a black-box GSE setting. The system comprises a retriever and a generator, whose internal mechanisms are inaccessible. The operator can only rely on external input–output observations, and is thus limited to optimizing existing content or introducing new contents into the external corpus.

2.3.2 Black-box Setting. The system is treated as a black box, where the internal parameters and decision processes of both retrieval and generation components are inaccessible to the G-SEO operator, although they may be indirectly influenced.

2.3.3 Content-based Interface. The G-SEO operator is restricted to modifying existing content source or introducing newly constructed data into the corpus, without altering the system implementation. The only observable feedback from the system is the generated response.

These assumptions provide a unified framework for analyzing G-SEO methods and highlight the core optimization challenges arising from the interaction between retrieval and generation processes.

3 Evaluation Framework

Due to the tight coupling between retrieval and generation processes in GSEs, the evaluation of G-SEO faces new challenges. Unlike traditional SEO, which primarily relies on ranking-based metrics for assessment, G-SEO requires a comprehensive evaluation framework that can simultaneously characterize retrieval performance and the influence of generated outputs. In this section, we systematically review existing studies from three aspects: evaluation metrics, benchmark datasets, and experimental protocols, and summarize the key characteristics and limitations of current evaluation practices.

3.1 Evaluation Metrics and Criteria

Although a growing body of research on G-SEO has emerged in recent years, a unified evaluation standard for performance assessment has not yet been established. Drawing inspiration from traditional SEO evaluation paradigms, and incorporating the characteristics of generative retrieval systems, this paper considers “content source visibility” as the core objective for measuring the effectiveness of G-SEO optimization. Based on the two key stages in generative search engines (GSEs), namely “retrieval” and “generation”, we categorize existing evaluation metrics into two main groups: **(i) retrieval-oriented visibility** and **(ii) generation-oriented visibility**. For clarity comparison, we summarize the aforementioned evaluation metrics in Table 1.

3.1.1 Retrieval-Oriented Visibility. Retrieval-oriented visibility is conceptually highly aligned with traditional SEO evaluation paradigms. It is primarily used to measure the ranking performance or selection probability of optimized content during the retrieval stage, thereby reflecting the impact of optimization methods on the underlying retrieval

Table 1. A structured summary of evaluation metrics for G-SEO.

Category	Metric Type	Representative Metric
Retrieval-Oriented	Direct Ranking Signals	Selection Frequency [50]
		Ranking Position [36, 69]
	Ranking Changes	Average Rank Improvement [3, 23, 35]
		Retrieval Success Probability [35, 81]
	Probabilistic & IR Metrics	Hit@1 [21]
		Mean Reciprocal Rank (MRR) [21]
		Promotion Success Rate [33]
		NDCG@K [17]
		Document Selection Rate [17]
		F1-score [9, 39, 83, 85]
		Retrieval Success / Failure Rate [8, 63, 77]
	G-SEO Specific	Adoption Rank Gain [53]
Generation-Oriented	Citation Statistics	Correct Answer Ratio [17]
		Target Citation Proportion [47, 67]
		Sentiment Score [67]
		Citation Success Probability [71]
		PAWC [1, 13, 75, 84]
		Attribute-based Evaluation Framework [64]
	LLM-based Perception	G-EVAL-based Evaluation Framework [1, 12, 13, 84]
		LLM-Based Semantic Evaluation [82]
	Stability & Risk	Positive Optimization Frequency [13]
		Bootstrap Confidence Interval [65]
		Worst Case Performance [84]
		Downside Risk [84]
		Win-Tie Rate [84]
		Influence Measurement [11]
	Success Presentation Probability	Attack Success Rate [9, 17, 32, 39, 67, 68, 73, 83, 85]
		Toxicity Efficiency Score [39]
Auxiliary Quality Metrics	Generation Quality	ROUGE-L [43, 82]
		BLEU [43, 82]
		CorpFaith [5]
		OrigFaith [5]
		TF-IDF [71]
		Embedding Similarity [71]
		Jaccard Similarity [71]
		Faithful Credit [11]
		Perplexity [43, 67, 69]
		GEU [18, 71, 75]

mechanism. Existing studies propose a variety of metrics from different perspectives, which can be broadly categorized into three types: *direct ranking signals*, *ranking changes*, and *probabilistic and statistical measures*.

First, a line of work characterizes visibility using direct ranking signals, typically derived from model outputs through straightforward statistics. For example, in generative recommendation settings, Nestaas et al.[50] measure visibility by counting the frequency with which target products are selected by generative models. Tang et al.[69] and Kumar et al.[36] further treat ranking position as a direct proxy for visibility, assuming that higher-ranked items correspond to higher exposure probability.

Second, some studies evaluate optimization effectiveness by measuring relative improvements in ranking before and after optimization. Du et al.[23], Kim et al.[35], and Bagga et al.[3] quantify performance by computing the average rank improvement of target content. In addition, Kim et al.[35] and Zhang et al.[81] introduce probability-based metrics similar to Hit Rate to capture the likelihood that target content is successfully retrieved.

Furthermore, a number of studies adopt standard evaluation metrics from the information retrieval (IR) literature to systematically characterize overall ranking quality, thereby providing a unified and comparable framework for assessing the impact of optimization methods on retrieval result distributions. For instance, Dhaka et al.[21] evaluate performance using Hit@1 and Mean Reciprocal Rank (MRR), while Jin et al.[33] propose Promotion Success Rate, which measures the probability of target content appearing within the Top- k results to capture visibility improvement. Choi et al.[17] further employ metrics such as Document Selection Rate and NDCG@K to quantify the visibility of distracting content, thereby evaluating the effectiveness of interference methods. In adversarial settings, RAG poisoning studies[9, 39, 83, 85] commonly adopt traditional retrieval metrics such as F1-score to measure the extent to which corpus manipulation disrupts retrieval behavior, indirectly reflecting the system’s sensitivity to optimization or attack strategies. In addition, some works take a more outcome-oriented perspective by directly evaluating attack effectiveness through retrieval success and failure rates[8, 63, 77], providing a more direct characterization of how attacks influence system behavior at the output level.

Moreover, to better capture characteristics of generative retrieval, some works propose task-specific metrics. For example, Puerto et al.[53] introduce adoption rank gain, which treats the ordering of citations in generated responses as a ranking signal, and measures visibility via the average rank change across different adoption rates as well as its corresponding AUC.

3.1.2 Generation-Oriented Visibility. Generation-oriented visibility primarily focuses on how content sources are presented within generated responses, particularly citation behaviors and their influence on the final answer. From a methodological perspective, studies on generation-oriented visibility can be systematically characterized along several complementary analytical dimensions. Intuitively, content with higher visibility is more likely to appear in generated outputs and to be perceived more prominently by users.

Although a large body of work has investigated citation evaluation in generated responses, including approaches such as mapping content attributes into a 16-pillar GEO scoring framework[37], the ALCE framework based on multi-dimensional quality assessment[26], and the principle-driven CiteEval[76], these methods primarily focus on citation quality (e.g., correctness, completeness, and credibility) rather than visibility itself. As a result, such metrics cannot directly reflect the impact of optimization methods on content exposure.

To address this issue, a growing body of work has begun to explicitly model citation visibility, proposing a diverse set of metrics from different perspectives and gradually forming an analytical paradigm that ranges from direct statistics to multi-dimensional modeling. A line of approaches relies on explicit signals in generated outputs to indirectly reflect the visibility of target content. For example, Choi et al.[17] measure the proportion of original correct answers in generated responses to indirectly capture the influence of distracting content on visibility. Building upon this, Mochizuki et al.[47] and Sui et al.[67] directly quantify visibility by computing the proportion of target content among all citations. Beyond purely statistical measures, some studies extend visibility modeling to semantic and affective dimensions of generated content. For instance, Sui et al.[67] introduce a sentiment scoring module to evaluate the emotional polarity of generated responses, thereby capturing the prominence and influence of emotionally charged content in generation outputs. Furthermore, Aggarwal et al.[1] propose the PAWC metric, which jointly considers citation position and the length of

cited text. This metric models visibility based on two key factors—positional priority and content volume—under the assumption that sources appearing earlier in the response and occupying larger textual spans exhibit higher visibility. It has since been widely adopted[13, 75, 84] and has become a de facto standard for evaluating citation influence in the generation stage. Building on this line of work, Tian et al.[71] further incorporate citation success probability into the evaluation framework, modeling visibility from a probabilistic perspective. Meanwhile, some studies approach visibility from the perspective of content attributes, providing an indirect modeling framework. Sharma[64] constructs a multi-dimensional evaluation scheme based on regular expressions, assessing content sources across six aspects, including statistical evidence, citation density, use of domain-specific terminology, authority of language expression, degree of structural organization, and content depth. This perspective offers complementary insights into how intrinsic content characteristics influence their visibility in generated outputs.

Beyond objective measurements, user perception of citations in generated responses also constitutes a key dimension of visibility. In addition to traditional human evaluation approaches[22, 29], recent studies increasingly leverage LLMs to simulate user perception processes. For instance, based on the G-EVAL framework[42], Aggarwal et al.[1] propose a fine-grained evaluation scheme across seven dimensions, namely relevance, influence, diversity, uniqueness, click-follow probability, positional salience, and content volume, to assess the impact of content sources in generated responses. This framework has been widely adopted in subsequent studies[84]. Chen et al.[12, 13] further extend this approach by introducing a unified evaluation pipeline template and normalized scoring procedures across dimensions, enabling a more systematic and comprehensive assessment. In addition, Zhang et al.[82] evaluate alignment between generated outputs and user intent from multiple aspects, including specificity, category adaptability, diversity, and coverage, thereby indirectly reflecting content source visibility in generation results.

Furthermore, some studies characterize visibility from the perspectives of statistical stability and risk, aiming to evaluate the robustness and consistency of optimization methods under different conditions. Chen et al.[13] use the frequency of positive visibility scores as a simple statistical measure of method applicability. Sielinski[65] models visibility as an underlying distribution and quantifies uncertainty via bootstrap confidence intervals. Zhou et al.[84] introduce Worst Case Performance [6], Downside Risk [45], and Win-Tie Rate [55] to evaluate visibility stability from a risk-control perspective. Chen et al.[11] further propose a comprehensive framework that captures visibility through both micro-level dimensions (Exposure, Faithful Credit, and Causal Impact) and macro-level dimensions (Mean Influence Level, Influence Coverage, and Influence Stability), systematically characterizing the influence of content sources in generated responses.

In adversarial generative retrieval settings, visibility is further reformulated as the probability of successfully presenting target content in generated outputs. Studies such as Zou et al.[85], Yang et al.[32], Li et al.[39], Chen et al.[9], Zhang et al. [83], Tan et al. [68], Choi et al. [17], Wang et al. [73], and Sui et al. [67] measure attack effectiveness via the probability that a target answer is generated, commonly referred to as attack success rate. Li et al. [39] further propose a Toxicity Efficiency Score to jointly characterize the trade-off between attack success rate and the impact on retrieval performance, providing a more comprehensive evaluation of optimization or attack strategies.

It is worth emphasizing that some studies also introduce generation-quality metrics, such as ROUGE-L and BLEU for evaluating factual accuracy of generated outputs[43, 82], CorpFaith and OrigFaith for assessing faithfulness to original content sources[5], TF-IDF, embedding similarity, and Jaccard similarity[71], Faithful Credit[11], perplexity for fluency evaluation[43, 67, 69], as well as GEU for evaluating overall generative retrieval quality[18, 71, 75]. Although these metrics are effective in characterizing generation quality, they do not directly correspond to the core objective of G-SEO, namely the improvement of content source visibility. In generative retrieval scenarios, even low-quality

or partially inaccurate content may still be retrieved and exposed by the system. Therefore, such metrics are more appropriately used as auxiliary evaluation criteria, particularly when distinguishing between methods with similar visibility performance, for instance by incorporating readability and credibility as secondary dimensions[11].

Finally, it is important to note that metrics based on generation-oriented visibility generally provide a more direct reflection of the practical effectiveness of G-SEO methods, as they are closely aligned with generative retrieval scenarios and end-user perception. However, retrieval-oriented visibility should not be overlooked. Users typically follow a top-down reading pattern when consuming generated responses, and RAG systems exhibit a similar sequential bias during information aggregation. Naphade[49] demonstrate through extensive controlled experiments that models tend to favor evidence presented earlier over information introduced later in the generation process. In other words, content sources appearing at the top of the retrieval ranking often receive higher attention weights. This positional bias implies that retrieval-stage ranking continues to play a critical role in determining overall visibility.

3.2 Datasets

As an emerging research area, G-SEO still lacks a systematic organization of datasets and benchmark suites. Existing studies typically construct or select datasets according to specific task requirements, resulting in fragmented data sources and heterogeneous design objectives. To address this issue, this section systematically reviews current G-SEO-related datasets and their characteristics, and further analyzes the applicability of different types of datasets to various task scenarios and research objectives.

3.2.1 G-SEO-Specific Benchmark Datasets. In existing studies, G-SEO-related datasets can be broadly understood from two perspectives: data construction methodology and application scenarios.

A portion of the literature tends to construct small-scale, task-specific datasets for preliminary validation and exploratory analysis. For example, Nestaas et al.[50] and Kumar et al.[36] build experimental datasets based on synthetic product webpages and product information to evaluate the effectiveness of generative manipulation methods; Zhang et al.[82] design a specialized dataset for multimodal recommendation scenarios; Dhaka et al.[21] construct a document embedding retrieval dataset containing multiple structured categories to assess optimization performance under complex document structures. In addition, some studies utilize existing large-scale dialogue datasets, such as WildChat-1M, for analysis[54], supporting the design of G-SEO strategies in specific vertical domains.

As research progresses, several works begin to construct more representative general-purpose benchmarks. Among them, GEO[1] is the first large-scale benchmark systematically introduced in the G-SEO domain, covering multi-domain generative retrieval tasks. It includes not only real user queries but also complex reasoning problems and synthetic queries, providing a unified evaluation foundation for subsequent studies. Building on this, later work extends GEO in terms of data diversity and evaluation complexity. For instance, Chen et al.[13] expand the query space by generating multiple semantic variants for each query, enabling evaluation of generalization ability under more complex retrieval conditions. Meanwhile, several datasets characterize G-SEO tasks from different modeling perspectives. The content-centric CC-GSEO-Bench[11] pairs content sources with query clusters to focus on evaluating the influence of content on generated outputs. MMIQ[71] introduces persona simulation and retrieval intent modeling to capture diverse user behaviors, further providing evaluation variants for complex structures and out-of-distribution scenarios.

For specific application domains, researchers have also developed a variety of domain-specific datasets. For example, Lüttgenau et al.[43] follow the GEO construction pipeline[1] to design a tourism-oriented dataset; Wu et al.[75] construct both real-world commercial query datasets and Researchy-GEO for open research scenarios; E-GEO[3] is the first

G-SEO dataset targeting e-commerce settings, containing multi-sentence consumer queries, product lists, and shopping context information; ProductBench[33] is built on Amazon search results and provides structured product information; Du et al.[23] further extend to a multimodal setting by constructing a G-SEO dataset that integrates textual and visual information. C-SEO[53] unifies question answering and product recommendation tasks within a conversational framework, providing a multi-domain evaluation environment across different scenarios.

It is worth noting that real-world generative retrieval tasks are considerably more complex than standard text-based datasets, often involving a mixture of structured and unstructured information. To better approximate practical applications, some studies directly collect raw web content as the data source. For instance, RAGDOLL[52] provides webpage data covering multiple product categories with rich attributes such as brand and model information; SAGEO[35] further preserves structured webpage elements (e.g., titles, meta descriptions, headings, and schema markup) to more realistically simulate optimized content sources. Tang et al.[69] combine the RAGDOLL dataset with synthetic product data from Kumar et al.[36] for experimental validation.

Finally, since G-SEO is inherently a competitive information retrieval problem, some studies adopt competitive search datasets with high task relevance. For example, Bardas et al.[5] employ two classical competitive search datasets[48, 56] for evaluation.

3.2.2 General Question Answering Datasets. Since generative retrieval systems are inherently designed for question answering (QA) scenarios, traditional QA datasets remain highly applicable and valuable for G-SEO research. This is particularly evident in RAG-based adversarial attack and data poisoning studies, where a substantial body of work directly leverages established QA benchmarks for experimental evaluation. From the perspective of dataset utilization, existing studies can be broadly categorized into two paradigms: single-dataset usage and multi-dataset combinations.

Specifically, Zhang et al.[81] conduct experiments on the MS MARCO dataset[4], while Cho et al.[16] perform systematic evaluations of perturbation effects using the Natural Questions (NQ) dataset[38], TriviaQA[34], and SQuAD[58]. In contrast, Shafran et al.[63] and Chen et al.[9] conduct joint experiments on both the Natural Questions (NQ) dataset[38] and the MS MARCO dataset[4]. Furthermore, Zhang et al. [83], Chaudhari et al.[8], Yang et al.[32], Li et al.[39], Sui et al. [67] and Zou et al.[85] jointly employ the Natural Questions (NQ) dataset[38], the MS MARCO dataset[4], and HotpotQA[79] to cover diverse task characteristics, including multi-hop reasoning and open-domain retrieval.

Building upon this, several studies further expand data sources to enhance the complexity and realism of evaluation scenarios, thereby enabling a more comprehensive assessment of model performance across diverse settings. For example, Chen et al.[14] not only utilize a subset of the MS MARCO dataset[4], but also incorporate data from PROCON.ORG¹ to simulate opinion manipulation in controversial public topics. Xue et al.[77] employ the NQ dataset[38], MS MARCO dataset[4], and SQuAD[58] for open-domain question answering evaluation, and further introduce the WikiASP dataset[28] to complement testing in generative task scenarios. Tan et al.[68] extend their evaluation by incorporating additional datasets such as FiQA[44] and Quora[70], thereby covering a broader range of query distributions. Similarly, Wang et al.[73] go beyond the NQ dataset[38] and FiQA[44] by introducing NFCorpus[7] and SciDocs[19], enabling the evaluation of cross-domain generalization performance of the proposed methods.

Moreover, since RAG systems serve as the upstream infrastructure of generative retrieval, RAG-oriented datasets can, in principle, be directly adapted for G-SEO tasks. For instance, Chen et al.[12] employ the MRAMG dataset[80] to validate the effectiveness of multimodal G-SEO methods.

¹<https://testing-www.procon.org/>

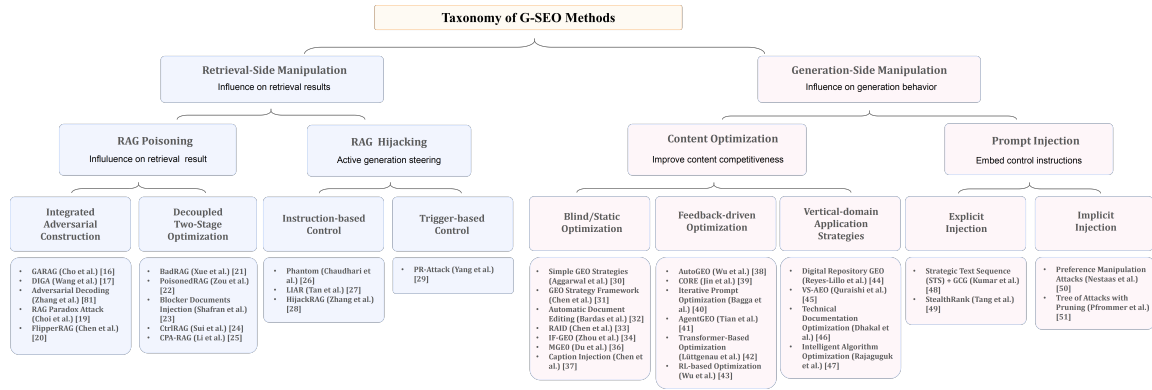


Fig. 3. Taxonomy of G-SEO Methods. This diagram classifies G-SEO methods based on their intervention point in the retrieval-generation pipeline. Retrieval-side manipulation methods (RAG Poisoning and RAG Hijacking) primarily alter the ranking of retrieved results, while generation-side methods (Content Optimization and Prompt Injection) focus on affecting the interpretation and integration of retrieved information during response generation. Each category is further subdivided into specific approaches and representative works.

More broadly, the strong alignment between generative retrieval and downstream applications implies that nearly any dataset containing a “question–candidate answer sources” structure can be leveraged for G-SEO research. In this sense, even G-SEO-specific datasets can be viewed as specialized extensions of QA datasets. It is important to note that, in the context of G-SEO, the effectiveness of QA datasets relies on a critical assumption: each question should be supported by multiple content sources. This is because both retrieval-oriented and generation-oriented visibility fundamentally aim to evaluate the contribution of specific content sources within the final generated answer. If an answer is derived from a single source, visibility analysis becomes meaningless due to the lack of comparative context.

3.3 Experimental Setting Protocols

Similar to most research in artificial intelligence, the evaluation of G-SEO methods should be conducted in a controlled experimental environment. Specifically, different optimization methods should be compared under fixed configurations of retrieval and generation components to ensure fairness, comparability, and reproducibility of experimental results. Given the inherent stochasticity of generative models, their outputs may exhibit significant variability under different sampling processes or prompt perturbations. Therefore, experiments are typically conducted over multiple independent runs or under varied prompting conditions, with average performance or statistical stability metrics reported to mitigate the influence of randomness.

Furthermore, unlike traditional retrieval or generation tasks, the primary objective of G-SEO lies in measuring *optimization effectiveness* rather than absolute performance. As such, evaluation metrics should emphasize relative improvements before and after optimization (i.e., post-optimization improvement), rather than reporting static performance scores alone. Concretely, it is essential to focus on changes in content visibility or influence-related metrics, thereby capturing the marginal gains introduced by optimization methods and more accurately reflecting the practical effectiveness of different G-SEO strategies.

4 Taxonomy of G-SEO Methods

As illustrated in Figure 3, we categorize existing G-SEO methods into two groups based on the structural characteristics of the retrieval-generation pipeline in GSEs: (i) **Retrieval-Side methods** and (ii) **Generation-Side methods**. This taxonomy is not based on the granularity of optimization targets (e.g., corpus-level versus individual content), but rather on the system workflow, distinguishing methods according to their intervention pathways on the core components—retrieval and generation—thereby establishing a direct correspondence between method types and system architecture. Specifically, **retrieval-side methods** intervene at the corpus level by introducing adversarial or manipulative content, thereby influencing the matching and ranking behavior of the retriever. Their objective is to increase the likelihood that target content is included in the retrieved candidate set. These methods do not directly act on the generative model; instead, they indirectly affect the final output by altering the distribution of retrieved content and, consequently, the information available for generation. In contrast, **generation-side methods** directly target the generation stage by optimizing the semantic representation, structural organization, or by embedding implicit instructions (i.e., prompt injection) within the content. These approaches influence how the model interprets and integrates input information, thereby increasing the likelihood that specific content is selected and more prominently expressed in the generated response. It is important to emphasize that this classification is fundamentally based on the point of intervention within the system rather than differences in content granularity. Although most methods primarily target a single stage, the tight coupling between retrieval and generation in black-box RAG systems implies that interventions at one stage often propagate through cascading effects to influence overall system behavior. Therefore, both the primary intervention pathway and potential cross-stage impacts should be considered when analyzing G-SEO methods.

Finally, we exclude backdoor attacks on RAG systems from the scope of G-SEO. For instance, some studies implant hidden triggers within models to manipulate generation behavior [15]. Such approaches rely on direct access to model parameters or system architecture, violating the black-box assumption adopted in this work. Therefore, they are not considered content-level G-SEO methods.

4.1 Retrieval-Side Manipulation

Prior work has shown that both the ranking and composition of retrieved content can significantly influence the outputs of RAG systems [49]. Compared to methods that directly manipulate the generation process, retrieval-side approaches build on this key property by focusing on increasing the likelihood that target content is selected during retrieval, thereby indirectly enhancing its visibility and influence in the final generated response. Existing methods typically achieve this by injecting carefully crafted adversarial or manipulative content into the corpus to intervene in the retrieval process. To further distinguish different mechanisms of intervention, we categorize retrieval-side manipulation methods into two subtypes based on their pathways of influence on the downstream generation process: **RAG Poisoning** and **RAG Hijacking**. **RAG Poisoning** primarily focuses on constructing content such that, once retrieved, it can naturally integrate into the generation context at the semantic level and directly influence the output. These methods rely on strong semantic alignment between the injected content and the query, passively guiding the model to produce target information. In contrast, **RAG Hijacking** introduces a stronger form of control by incorporating instruction-like structures or signals into the retrieved content. These signals are designed to steer the model’s attention allocation or reasoning process during generation, thereby biasing the output toward specific information. Although such methods still operate through the retrieval stage, their influence mechanism is more akin to cross-stage manipulation, exerting

more directed control over the generation process. It is important to emphasize that both categories fundamentally fall under data poisoning attacks. Their distinction mainly lies in the manner and strength of their influence on the generation stage, rather than indicating two inherently distinct attack paradigms.

4.1.1 RAG Poisoning. RAG Poisoning is one of the most rapidly evolving research directions in G-SEO. Its core idea is to inject carefully crafted manipulative content into the corpus so that it is preferentially retrieved and subsequently incorporated into the generation context, thereby influencing the final output and potentially inducing misleading or incorrect responses. From a mechanistic perspective, these methods can be understood as data-level interventions: they manipulate retrieval ranking signals to reshape the retrieval distribution and indirectly constrain the input conditions of the generation stage, rather than directly altering itself. Despite variations in implementation, such approaches generally share three key characteristics: (1) reliance on corpus-level content injection; (2) influencing retrieval preferences through semantic structuring, textual organization, or implicit instruction embedding; and (3) treating retrieval visibility as the primary mediating factor for affecting generation outcomes. Based on differences in optimization strategies and modeling paradigms, existing RAG Poisoning methods can be broadly categorized into end-to-end integrated adversarial construction and two-stage decoupled optimization.

End-to-end adversarial construction methods typically adopt unified optimization frameworks that jointly model retrieval visibility and generation influence. Technically, these approaches span multiple paradigms, including evolutionary search, constrained generation, and surrogate-based optimization. Cho et al. [16] propose GRAG (Genetic Retrieval-Augmented Generation), which leverages genetic algorithm-driven perturbations to enhance adversarial properties in both retrieval and generation. Wang et al. [73] introduce DIGA (Dynamic Importance-Guided Genetic Algorithm), combining token importance modeling with semantic similarity optimization to improve retrieval ranking while preserving relevance. Zhang et al. [81] incorporate retrieval-aware constraints into generation to optimize adversarial text construction. Choi et al. [17] reconstruct manipulative information into high-credibility expressions via trusted source simulation, increasing its likelihood of being retrieved and adopted. Chen et al. [14] propose FlipperRAG, which models ranking preferences via a surrogate retriever and optimizes embedded adversarial signals through trigger-based mechanisms. Overall, these methods achieve high attack efficiency and strong local controllability, but are often sensitive to retrieval model distributions and ranking strategies, limiting cross-system generalization.

In contrast, two-stage optimization methods explicitly decouple retrieval and generation objectives, optimizing them separately and combining them at the content level. Compared to integrated approaches, these methods emphasize cross-query generalization and structural transferability, leading to greater adaptability, albeit often requiring access to or estimation of target queries. Xue et al. [77] propose BadRAG, which leverages topic-level trigger modeling and contrastive optimization to enhance robustness and generalization. Zou et al. [85] introduce PoisonedRAG, which decomposes manipulative content into retrieval-enhancing and generation-guiding components, enabling coordinated optimization. This simple yet effective design has been widely adopted in subsequent work. For example, Shafran et al. [63] employ hill-climbing search with textual perturbations to induce target outputs; Sui et al. [67] propose CtrlRAG, which generates manipulative content via contextual reconstruction and role-based framing, combined with mask-based perturbation and ranking-guided search for stable control in black-box settings; Li et al. [39] present PCA-RAG, integrating prompt-based adversarial generation, multi-model optimization, and retrieval-aware evaluation to construct high-quality adversarial texts. Compared to integrated methods, these approaches offer greater structural flexibility and generalization, but involve more complex optimization and rely more heavily on query assumptions.

Overall, end-to-end methods tend to achieve higher single-instance attack efficiency with lower construction complexity but suffer from limited generalization, whereas two-stage methods provide stronger generalization and transferability at the cost of increased query dependence and optimization overhead. This line of work fundamentally reflects a trade-off among attack strength and query dependency.

4.1.2 RAG Hijacking. RAG Hijacking also falls under retrieval-side manipulation, where the attack relies on injecting crafted manipulative content into the corpus so that it can be retrieved and subsequently incorporated into the generation process. Unlike RAG Poisoning, which primarily affects generation outcomes in a passive manner by altering retrieval rankings, RAG Hijacking emphasizes actively intervening in the model’s reasoning process and output behavior after the manipulated content has been included in the generation context. Therefore, the key of this line of work lies not only in improving the retrievability of the injected content, but more importantly in how to hijack and steer the model behavior during the generation stage. Specifically, such methods typically consist of three core components: (1) retrieval interference text, which increases the likelihood that the manipulated content is retrieved; (2) hijacking text, which encodes the control intent and often contains specific structural or semantic patterns to influence attention allocation or reasoning pathways; and (3) control instructions, which trigger or enforce the desired generation behavior. Based on these components, existing methods can be broadly categorized into two types of control strategies.

One line of work explicitly constructs instruction-bearing content, allowing control signals to enter the generation process in a directly interpretable form. These methods typically adopt structured concatenation or joint optimization strategies, combining retrieval-relevant text with manipulation instructions within the same context so that the attack intent is explicitly expressed during generation. Chaudhari et al. [8] propose Phantom, which constructs retrieval interference text and generation hijacking text using strategies such as HotFlip and MCG, respectively, and concatenates them with control instructions to jointly induce diverse adversarial behaviors. Tan et al. [68] introduce LIAR, which employs gradient-based token replacement to increase the influence of manipulated text during generation. Zhang et al. [83] leverage the HackAPrompt dataset [62] to select and reconstruct prompt templates, transforming queries into control targets and explicitly injecting hijacking intent into the generation stage. Overall, these methods provide strong controllability; however, their explicit instruction structures make them more susceptible to detection and filtering.

Another line of work avoids exposing explicit instructions and instead constructs implicit triggering patterns that subtly alter model behavior during generation. These methods typically introduce fixed or learnable triggers that are activated under specific contextual conditions, thereby influencing the model’s generation tendencies. Yang et al. [32] propose PRAttack, which employs a fixed trigger pattern and jointly models soft prompts and the distribution of toxic text via alternating optimization, and they use a zeroth-order optimization scheme to update the probability distribution of the toxic text, enabling the construction of a stealthy triggering mechanism that induces biased outputs while preserving surface-level semantic consistency. Compared to explicit approaches, these methods offer improved stealthiness and cross-scenario adaptability, but rely more heavily on optimization strategies and provide less fine-grained control over generation outcomes.

RAG Hijacking does not aim to change what the model retrieves, but also how the model utilizes the retrieved information. Explicit approaches achieve strong control by directly interfering with instruction parsing, whereas implicit approaches provide more covert guidance through latent triggering mechanisms. The core trade-off of this line of work lies in balancing control strength, stealthiness, and optimization complexity.

Overall, retrieval-side manipulation methods exploit the fundamental dependency of RAG systems on retrieved context, effectively shifting the attack surface from the generation module to the data and retrieval pipeline. By

intervening in the selection and composition of retrieved content, these methods reshape the input distribution of the generator, thereby indirectly but effectively influencing downstream outputs. Despite sharing a common data-centric paradigm, RAG Poisoning and RAG Hijacking differ in their mechanisms of influence: the former primarily relies on semantic alignment to passively bias generation, while the latter introduces structured or instruction-like signals to actively steer model behavior. This distinction reflects a broader spectrum of control, ranging from implicit distributional shifts to explicit behavioral intervention. From a more general perspective, retrieval-side manipulation methods are not driven by a single optimization objective, but instead operate under a multi-dimensional trade-off among attack effectiveness, controllability, stealthiness, and cross-scenario adaptability. Characterizing this trade-off structure is essential for systematically understanding the operational boundaries of different attack paradigms.

4.2 Generation-Side Manipulation

From the perspective of the operational mechanism of GSEs, generation-side manipulation methods directly target the generation stage by influencing how the model interprets, integrates, and expresses input information, thereby altering the final output. Existing studies typically focus on improving the visibility and influence of target content in generated responses without explicitly intervening in the retrieval results. As a result, the retrieval process is often treated as a fixed input condition rather than being explicitly modeled. Based on their technical pathways and intervention strategies, generation-side methods can be categorized into **Content Optimization** and **Prompt Injection**. The key distinction lies in whether semantic information directly associated with the content source is explicitly used to reconstruct or manipulate the original content. The former emphasizes improving content representation to enhance its competitiveness during generation, while the latter focuses on introducing control signals to impose more direct intervention on the generation mechanism. **Content Optimization** methods refine the content itself through semantic rewriting, structural reorganization, or information enrichment, making it more likely to be incorporated and assigned higher importance during the model's information fusion process. As a result, such content achieves greater visibility in the final generated output. In contrast, **Prompt Injection** methods exploit the model's sensitivity to instruction-like inputs by embedding implicit or explicit control signals into the context, thereby directly influencing the model's reasoning pathways and attention allocation. This allows the model to be steered toward following specific instructions to produce targeted outputs during generation. In practice, the boundary between the two is often blurred: content optimization may incorporate lightweight directive signals, while prompt injection may rely on semantic restructuring to improve stealth and effectiveness. Therefore, analyzing generation-side manipulation requires jointly considering its dominant intervention pathway and the interplay between semantic modeling and instruction-level control.

4.2.1 Content Optimization. Content optimization methods are typically realized through direct modifications of the content itself. The core idea is to introduce semantically relevant augmentation signals while preserving the original semantic intent, thereby increasing the likelihood that the content is attended to and adopted by generative retrieval models. In essence, these methods enhance the competitiveness of content during the generation process through rewriting, restructuring, or semantic enrichment. From the perspective of information availability and optimization conditions, existing studies exhibit varying degrees of reliance on feedback signals. Some methods operate without access to feedback and rely solely on static inputs for content rewriting, which we refer to as the blind optimization setting. In contrast, other methods leverage limited or indirect feedback signals to perform iterative optimization, progressively aligning content with the desired generation behavior.

Under the blind optimization setting, existing approaches cannot access internal ranking, citation, or response signals of GSEs and rely entirely on static inputs for content modification. Aggarwal et al. [1] first propose eight fundamental LLM-based content modification strategies, enhancing textual expressiveness along multiple dimensions such as tone, structure, and credibility to improve visibility in generated responses. Chen et al. [10] further introduce a strategic GEO framework at the system level, emphasizing the construction of high-authority external link structures to increase the probability of being cited by generative systems. Bardas et al. [5] propose a prompt-based content optimization paradigm, where optimization objectives and historical ranking signals are structurally encoded to guide the rewriting process in a controlled manner.

Building upon these static rewriting strategies, subsequent work explores the incorporation of explicit search intent modeling to improve the alignment between optimized content and potential user needs. Chen et al. [13] introduce RAID, which extracts search intent from the content source and incorporates a multi-role reflection mechanism to generalize intent representations, enabling the rewritten content to maintain stronger semantic alignment across diverse query contexts. Zhou et al. [84] develop IF-GEO, which generates multiple intent-aware optimization suggestions and integrates them through a conflict-aware fusion mechanism to produce a unified optimization strategy. Zhang et al. [82] present Pinterest GEO, modeling potential queries through reverse search and visual-language signals while constructing link structures that reinforce authority propagation. Spriestersbach et al. [66] do not directly introduce a G-SEO method, but their GIO intent classification framework highlights the fundamental role of search intent in generative answer construction.

In contrast to methods that do not rely on system feedback, another line of research leverages limited or indirect feedback signals to perform iterative optimization by analyzing the responses of generative systems. Wu et al. [75] develop AutoGEO, which characterizes citation preferences and visibility disparities, distills them into actionable insights, and organizes them into high-quality rule sets for content rewriting. Jin et al. [33] formulate CORE, incorporating proxy models and feedback-driven optimization to derive signals from discrepancies between observed and target rankings, thereby enabling iterative alignment between content and desired ranking outcomes. Bagga et al. [3] build a prompt meta-optimization framework that continuously refines a set of foundational optimization prompts through performance-driven self-reflection, improving both robustness and generalization. Tian et al. [71] focus on citation failure diagnosis and implement AgentGEO, which identifies content-level issues using a taxonomy of citation errors and applies memory-driven strategy selection to iteratively repair them. Beyond feedback-driven optimization, some studies explore learning-based approaches at the model level. Lüttgenau et al. [43] model the transformation between original and optimized content through supervised learning. Wu et al. [75] further incorporate reinforcement learning to enable end-to-end learning of content optimization capabilities.

Research has also extended to multimodal content optimization. Du et al. [23] propose MGEO, which introduces a gradient-based StealthRank [69] framework to generate adversarial textual suffixes and applies Projected Gradient Descent to construct image perturbations. By jointly optimizing ranking constraints and cross-modal consistency, the framework enhances the alignment between textual and visual signals. Chen et al. [12] improve visibility in multimodal scenario by rewriting image captions to transform visual semantics into more retrieval-friendly textual representations.

At the application level, several studies develop systematic G-SEO frameworks for specific vertical domains. Reyes-Lillo et al. [59] propose a GEO strategy for institutional repositories, focusing on semantic structure, metadata design, and system interoperability. Quraishi et al. [54] analyze local service scenarios and design customized optimization strategies for different service types. Dhakal et al. [21] compare semantic organization and structured representations in RAG systems, highlighting the importance of natural language structuring. Rajaguguk et al. [57] propose a comprehensive

optimization framework for AI search across diverse dimensions, covering content structure, accessibility, query adaptation, and system monitoring.

Overall, most LLM-based rewriting approaches tend to outperform traditional SEO strategies in experimental settings, suggesting that conventional SEO paradigms are not fully adapted to generative retrieval environments and highlighting the need for G-SEO. However, conflicting findings have also been reported. Sharma et al. [64] and Puerto et al. [53] report that traditional SEO methods may outperform G-SEO under certain conditions. Puerto et al. [53] integrate eight LLM-based rewriting methods within the C-SEO benchmark and incorporate a Markdown summarization guiding strategy, observing that traditional SEO performs better across eleven methods. Sharma et al. [64] further validate this observation through large-scale real-world product recommendation experiments. We attribute these discrepancies to two main factors. The first concerns evaluation design, where both studies rely on visibility metrics based on pattern matching and ranking shifts, which are more aligned with traditional ranking-oriented optimization and do not fully capture the answer generation process, thereby favoring traditional SEO. The second relates to task settings, as different application scenarios vary in their sensitivity to optimization strategies. In particular, product recommendation tasks rely more heavily on ranking signals, making traditional SEO more competitive in such settings.

Collectively, content optimization research has evolved from heuristic methods based on static rewriting toward more complex paradigms that incorporate feedback-driven learning and multimodal coordination, reflecting a shift from simple expression enhancement to alignment with generative behaviors.

4.2.2 Prompt Injection. In contrast to content optimization methods that directly modify the content itself, prompt injection approaches operate by embedding specially crafted instructions or control-oriented text into the content source, with the goal of explicitly influencing the reasoning process and decision-making behavior of large language models during generation. The key idea lies in exploiting the sensitivity of LLMs to input context, allowing injected text to be interpreted as task-relevant signals and thereby prioritized in the generation process. From a methodological perspective, existing work can be broadly categorized into two directions. One line of research formulates prompt injection as an optimization problem, where adversarial prompts are generated to directly manipulate ranking or recommendation outcomes. The other line focuses on structured reasoning and strategy search, constructing injections with improved stealthiness and generalization.

For optimization-based prompt generation, prior work typically defines objective functions and constraints to guide the construction of injection text that maximizes the visibility or ranking of target content. Kumar et al. [36] further introduce a gradient-based approach, where strategic text sequences are iteratively refined using the GCG algorithm by replacing tokens with high-gradient alternatives under a cross-entropy objective, effectively improving the ranking of target products. Tang et al. [69] develop StealthRank within the COLD decoding framework, constructing an energy-based objective augmented with stealth constraints, and employ Langevin dynamics to generate injection prompts that are both effective and difficult to detect.

Beyond explicit optimization, another line of work explores prompt construction from the perspective of reasoning and structured search, aiming to improve robustness and adaptability. Netaas et al. [50] demonstrate that carefully designed injected prompts can manipulate LLM preferences, leading to the preferential recommendation of injected, fictitious web pages, and highlight the generality of such attacks across settings. Pfrommer et al. [52] introduce the Tree of Attacks with Pruning framework, which models prompt construction as a tree search process. Each node incorporates meta information such as historical injections, model responses, and ranking feedback, and is expanded via chain-of-thought reasoning to generate new attack candidates. Prompts are then evaluated and pruned based on

their effectiveness in improving ranking across multiple recommenders. Xue et al. [77] exploit alignment properties of generative models with human values by designing semantically guided injections, such as prompts emphasizing privacy or safety, to induce refusal behaviors and indirectly control system outputs.

Prompt injection methods offer the advantage of influencing generation behavior without altering the original content semantics, enabling efficient and transferable manipulation at the input level. However, their effectiveness heavily depends on the model’s sensitivity to injected instructions, leading to variability across different systems. Moreover, with the emergence of defense mechanisms such as instruction filtering and context isolation, the robustness and stealthiness of such approaches are increasingly challenged.

In summary, generation-side manipulation methods operate at the final stage of GSEs, where heterogeneous inputs are integrated and transformed into coherent natural language outputs. Unlike approaches that target the retrieval stage, these methods do not directly alter the source or distribution of input content, but instead focus on how information is organized, expressed, and presented during generation. Content optimization and prompt injection constitute two primary forms of intervention. Content optimization reshapes content representations by enhancing semantic salience and structural clarity, thereby influencing the likelihood that certain information is selected, integrated, and emphasized during generation. Prompt injection, in contrast, embeds external control signals into the input context, guiding the model to follow specific instruction patterns and exerting more direct control over output behavior. The key distinction lies in whether control is achieved through representation shaping or through instruction-driven signaling. From a broader perspective, generation-side manipulation forms a spectrum of intervention granularity, ranging from implicit modulation of content importance to explicit steering of generation behavior, reflecting the diversity and layered nature of control mechanisms in this space.

5 Challenges and Discussions

Although G-SEO has attracted increasing attention from the research community, it remains in an early stage of development. Despite a growing body of exploratory work, the overall improvement achieved by existing methods remains relatively limited. We argue that this limitation does not primarily stem from deficiencies in individual method design, but rather from inherent constraints and complexities in generative retrieval tasks. In particular, the black-box nature of GSEs introduces substantial uncertainty, limiting access to reliable feedback signals, while the lack of unified and objective evaluation frameworks further hinders systematic comparison and iterative improvement. Against this backdrop, we provide a structured discussion of key challenges and future directions in G-SEO, grounded in the evolution of existing methods and their limitations. In addition, we consider defensive mechanisms as well as broader ethical and regulatory implications, aiming to characterize the boundaries and research space of G-SEO from a more comprehensive perspective.

5.1 Challenges

The task setting of black-box GSEs presents a fundamental challenge for current G-SEO research. In this context, the high opacity of the retrieval and generation components within GSEs places content providers at a significant disadvantage in terms of information access and feedback perception. This asymmetry manifests in practice as an implicit adversarial relationship between the content source and the generated response, leading to high uncertainty and uncontrollability in the optimization process. At the same time, the majority of G-SEO methods rely on reconstructing original content or generating new content sources, inevitably introducing semantic shifts in the process. It is important to emphasize that these semantic shifts are different from entity-level mapping shifts [27], as they fundamentally involve

fine-grained perturbations in the attention distribution of the generation model. Even minor expression changes, which might appear insignificant from a human perspective, can lead to significant differences in the generated output within black-box GSEs. Thus, how to controllably characterize and leverage this attention sensitivity in the absence of internal observability becomes a core challenge in G-SEO optimization. Although some studies have attempted to mitigate these issues by constructing proxy models [14, 33, 75], leading to some improvement in optimization performance, such approaches remain constrained by theoretical limits. On one hand, the retrieval ranking mechanism in real GSEs is difficult to replicate precisely; on the other hand, the decision-making process of the generation module in content selection and information integration is also unobservable. Therefore, optimization strategies based on proxy models inevitably face performance biases when transferred to real system simulation.

In addition to modeling challenges, the lack of a comprehensive and objective evaluation framework also constitutes a key bottleneck in the development of G-SEO. Most existing evaluation methods measure content visibility from a single perspective—either the generation side or the retrieval side, lacking a unified evaluation framework that simultaneously characterizes the dual capabilities of content sources as both retrieved and generated. Furthermore, the attribution issue in generative systems [40, 41, 51] further complicates the evaluation process. Inaccurate attribution mechanisms can obscure the true contribution of content, thereby interfering with objective assessments of optimization effectiveness.

5.2 Future Work Discussion

The core objective of G-SEO is to influence the implicit reasoning pathways of GSEs without directly modifying their internal parameters. This is achieved by performing targeted semantic restructuring and expressive refinement of specific content sources, thereby intervening in the model’s semantic interpretation process and increasing the likelihood of being prioritized during information integration and attention allocation in the generation stage. In essence, this process can be understood as a form of competition between content sources and the generation mechanism over semantic control and attention resources. Based on a systematic review and analysis of existing studies, we observe that although current G-SEO methods have made initial progress along multiple directions, the field as a whole remains in an exploratory stage, with its methodological framework and theoretical boundaries not yet fully established. Under this context, future research should move toward more systematic and multi-dimensional perspectives to further expand the capabilities and scope of G-SEO. Specifically, we identify several promising research directions that warrant focused investigation in the near future.

5.2.1 Long-Context G-SEO. Long-context modeling has long been a fundamental challenge in the field of Natural Language Processing (NLP) and is particularly critical for G-SEO. As GSEs built upon LLMs continue to expand their context windows, their behavior in processing long sequences will directly influence the effectiveness of optimization strategies. Most existing G-SEO methods are designed for relatively short-text scenarios, and their optimization signals may gradually weaken in long-context settings due to the dispersion of attention, thereby reducing overall effectiveness. In addition, although a substantial body of work adopts RAG-based architectures to approximate GSEs, there exist fundamental differences in input characteristics. In practice, GSEs typically operate over structurally complex web content, which includes not only textual information but also navigational elements, redundant descriptions, and cross-module signals. This leads to input sequences that are significantly longer and more structurally complex than standard textual corpora. Under such conditions, the likelihood of redundancy and semantic conflicts within long contexts increases, further diminishing the influence of optimized content during the information integration process. Therefore, future G-SEO research should move beyond surface-level content refinement toward deeper semantic

interaction modeling. This involves exploring mechanisms that maintain the perceptibility and prioritization of target content under long-context conditions, such as hierarchical semantic modeling, cross-paragraph attention guidance, and selective context amplification.

5.2.2 Blind G-SEO. The blind optimization setting, briefly introduced earlier, refers to optimizing content in the absence of any feedback from GSEs or external environments. This setting is not only of theoretical interest but also highly relevant in practice, where obtaining stable and high-quality feedback signals is often limited or infeasible. As such, designing effective optimization strategies under severely constrained information conditions represents a key research direction in G-SEO. In this context, the central challenge lies in establishing a clear and reliable optimization guidance that enables the optimized content to better align with the latent semantic preferences of GSEs as well as potential user queries, thereby maximizing optimization effectiveness. Some existing studies attempt to leverage search intent as a guiding signal for optimization [13, 84], which partially mitigates the uncertainty inherent in feedback-free settings. However, such approaches are inherently dependent on the accuracy of intent modeling, and misalignment in intent extraction may adversely affect both content quality and optimization outcomes. Future research may advance along two complementary directions. On one hand, developing more precise and robust methods for search intent modeling and alignment remains essential. On the other hand, alternative forms of optimization guidance could be explored, such as leveraging semantic distribution estimation or modeling latent user behavior, to improve robustness and generalization in blind optimization scenarios.

5.2.3 Cross-system G-SEO. As the number of GSEs continues to grow, designing specific G-SEO strategies for different GSEs can indeed achieve better optimization results. However, this process is often associated with high time costs and computational overhead. The main challenge lies in the need to deeply understand and adapt to each system’s semantic preferences and work mechanisms to ensure that the optimization strategy performs optimally. As GSEs continue to expand and diversify, the limitations of optimization strategies designed for individual systems have become increasingly apparent. Therefore, we propose cross-system G-SEO methods as a promising future research direction. The core idea is to design a lightweight intermediate component that allows optimization strategies to be transferred and adapted across different GSE systems, providing a more flexible and efficient solution to the ever-changing generative search environment. Specifically, future research could explore the following key areas: first, how to ensure the efficiency and consistency of optimization strategies when migrating across systems, possibly through shared knowledge bases, semantic alignment mechanisms, or standardization of intermediate representations; second, how to design adaptive algorithms that can accommodate the differences in generative preferences and semantic understanding across various systems.

5.2.4 SEO-compatible G-SEO. As highlighted by Puerto et al. [53], existing G-SEO methods do not consistently outperform traditional SEO across all generative retrieval scenarios. We attribute this phenomenon to differences in task characteristics and optimization objectives. In particular, tasks such as product recommendation are inherently closer to ranking problems, where the objective aligns more closely with ranking mechanisms in traditional search engines. In such settings, traditional SEO strategies remain highly effective, whereas current G-SEO approaches, which primarily focus on information integration and expression optimization during the generation stage, may exhibit comparatively limited performance. Based on this observation, developing SEO-compatible G-SEO methods emerges as an important research direction. It is necessary to establish a unified modeling framework that captures both generative and traditional retrieval paradigms, characterizing their relationships and differences in information

retrieval, ranking, and generation processes. Besides, future work should explore optimization strategies that can simultaneously accommodate ranking-oriented and generation-oriented objectives, thereby achieving more stable and consistent performance across diverse retrieval scenarios. Additionally, integrating structured signals from traditional SEO with the semantic modeling capabilities of G-SEO may offer a promising pathway toward a unified cross-paradigm optimization framework.

5.2.5 Anti Prisoner’s Dilemma G-SEO. Researchers have pointed out that in generative search environments, as an increasing number of content sources adopt G-SEO for optimization, a Prisoner’s Dilemma phenomenon may emerge [50]. Specifically, when all content sources employ G-SEO strategies simultaneously, the visibility of each individual source may be suppressed to some extent. In other words, the widespread adoption of G-SEO can induce a negative feedback effect that ultimately limits the effectiveness of optimization strategies. However, most existing G-SEO studies are conducted under the assumption of optimizing a single content source. In multi-party competitive settings, such methods often struggle to accommodate the dynamic interactions introduced by concurrent optimization efforts. In particular, when multiple content sources are optimized simultaneously, preventing mutual interference or cancellation of optimization effects becomes a critical challenge. Therefore, future research should shift toward G-SEO under competitive environments, with an emphasis on mitigating the Prisoner’s Dilemma. Promising directions include the incorporation of multi-agent game-theoretic modeling, collaborative or equilibrium-aware optimization strategies, and adaptive mechanisms that can dynamically respond to the evolving competitive landscape, thereby improving the stability and effectiveness of G-SEO in practice.

5.2.6 Anti-G-SEO and Defensive Mechanisms. In addition to research on G-SEO-based content optimization and manipulation strategies, there exists a complementary line of work, which we refer to Anti-G-SEO. From a fundamental perspective, G-SEO can be viewed as a form of content enhancement, aiming to improve the visibility of target information in generative outputs without modifying the underlying model parameters. However, this process often involves varying degrees of semantic restructuring and information shift, which in some cases may introduce inaccuracy or overly strategic expressions, thereby affecting users’ perception of information authenticity and reliability. From the perspective of user trust and system integrity, such optimization-driven enhancement is not always desirable. This motivates a natural research question: how to detect and suppress optimization-induced artifacts in order to recover or approximate the original semantic state of the content. Based on this perspective, the core objective of Anti-G-SEO can be understood as adversarially removing optimization bias, ensuring that GSEs present information in a manner that is more faithful to its intrinsic semantics rather than its strategically rewritten form. From a methodological standpoint, this direction can be further extended to a variety of defensive mechanisms, such like semantic consistency-based content purification. We argue that such approaches not only enhance user trust and interpretability of generative outputs, but also form an adversarial co-evolutionary relationship with G-SEO, thereby promoting a more balanced ecosystem between optimization and constraint in GSEs.

G-SEO, as an emerging technology, faces a range of challenges and research opportunities. While significant progress has been made, the field remains in its exploratory stage, with its methodological framework and theoretical foundations still under development. Future research should take a more systematic and multi-dimensional approach, addressing how to enhance the effectiveness and adaptability of G-SEO in various scenarios and fine-grained task settings. With continued exploration, G-SEO is poised to make broader contributions and innovations in the generative search space.

5.3 Ethical, Legal, and Regulatory Implications

The development of G-SEO not only introduces methodological and technical challenges, but also raises a series of important ethical, legal, and regulatory concerns. Fundamentally, while such approaches aim to enhance content visibility, they may simultaneously exert far-reaching effects on information fairness, user trust, and the stability of GSE ecosystems. From an ethical perspective, excessive or unconstrained optimization can lead to systematic biases in information presentation, causing users to encounter content that no longer strictly reflects true relevance, but rather the outcome of strategic intervention. For instance, prior studies have shown that authoritative information sources can significantly influence the generation behavior and citation tendencies of LLMs [10, 61, 72]. We argue that this phenomenon partly stems from the increasing reliance of LLMs on trusted sources during generation, which is originally intended to mitigate hallucination and improve output reliability. However, this same mechanism also creates opportunities for G-SEO, enabling existing approaches to enhance content visibility through the design of structured authority signals [1, 10]. Without appropriate constraints, such strategies may further amplify imbalances in information distribution and even introduce risks of disseminating misleading content. From a legal and regulatory standpoint, requirements related to information manipulation, advertising transparency, and algorithmic accountability are becoming increasingly stringent across different jurisdictions. As a result, the practical deployment of G-SEO methods faces more complex compliance challenges. In this context, future research should seek a balance between technological innovation and regulatory governance. On the one hand, it is necessary to promote transparency and auditability in G-SEO processes, ensuring that optimization behaviors are traceable and evaluable. On the other hand, establishing clear normative frameworks that define acceptable boundaries of optimization and allocate responsibilities across different application scenarios is equally critical, thereby fostering the long-term, sustainable development of the GSE ecosystems.

6 Conclusion

Overall, this survey provides a systematic review of G-SEO by establishing a unified analytical framework that covers research scenarios, task definitions, evaluation protocols, underlying mechanisms, and methodological paradigms. Existing studies show that G-SEO can influence the information integration and response generation processes of GSEs without directly modifying parameters, primarily through fine-grained adjustments to content semantics and expression. Furthermore, by examining G-SEO from both generation-side and retrieval-side perspectives, this work offers a coherent characterization of its evaluation principles and operational mechanisms, thereby enabling a more integrated understanding of different approaches.

Despite encouraging progress along multiple directions, G-SEO as a whole remains at an early stage of development. Current methods still exhibit notable limitations in terms of optimization effectiveness, theoretical grounding, and cross-scenario generalization. Building upon this observation, we summarize the key challenges facing the field and outline several promising avenues for future research. Moving forward, advancing G-SEO will require sustained efforts in improving modeling fidelity, developing comprehensive evaluation frameworks, enhancing cross-system generalization, and incorporating safety and regulatory considerations. Such efforts are essential for transitioning G-SEO from heuristic-driven practices toward a more systematic and interpretable paradigm, ultimately supporting the robust and sustainable evolution of generative search ecosystems.

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